

IT3091 – Bank Marketing

Model/Method Comparison Log

Model	Role	Key Settings	Why Included
Logistic Regression	Baseline	class_weight="balanced", max_iter=1000	Simple, coefficient-interpretable — a bank compliance team could read and justify individual predictions
Decision Tree	Alternative 1	class_weight="balanced", max_depth=8	Human-readable (traceable rules), tests whether a single tree captures enough non-linearity
Random Forest	Alternative 2	n_estimators=300, class_weight="balanced", max_depth=12	Ensemble expected to improve on both baseline and single tree; provides feature importances
XGBoost	Alternative 3	n_estimators=300, max_depth=6, learning_rate=0.1, scale_pos_weight set to the train class ratio	Gradient boosting, typically strongest raw performance; handles imbalance via weighting rather than resampling

Performance comparison (test set, n = 9,043)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Lift @ Top 20%
Logistic Regression	0.7552	0.2669	0.6257	0.3742	0.7722	0.4092	33.1%
Decision Tree	0.8326	0.3594	0.5510	0.4351	0.7561	0.3781	33.4%
Random Forest	0.8363	0.3689	0.5614	0.4453	0.7927	0.4211	34.5%
XGBoost	0.8267	0.3552	0.5898	0.4433	0.7824	0.4311	34.8%

(Base subscription rate in test set: 11.7% — all four models roughly triple this rate when targeting the top-scored 20% of clients.)

Interpretation and critical judgement

- No single model dominates every metric — the "best" model depends on what the bank is optimizing for. Under threshold-based metrics (accuracy, F1, ROC-AUC), Random Forest performs best. Under PR-AUC and Lift@20% — the metrics most aligned with a ranking/prioritization decision under severe class imbalance — XGBoost is marginally ahead. This is treated as a genuine finding, not noise: PR-AUC and Lift@Top-K are prioritized as the more decision-relevant metrics, since the actual business task (rank clients for limited call capacity) is a ranking problem, not a balanced classification problem.
- Decision Tree is dominated by both ensembles on nearly every metric, including the lowest PR-AUC (0.3781) and ROC-AUC (0.7561). Retained in the comparison as the "fully interpretable" reference point, not as a leading candidate.
- Logistic Regression achieves the highest recall (0.6257) but the lowest precision (0.2669) among the three models — a direct consequence of class weighting shifting the decision boundary toward sensitivity. This trade-off (catch more subscribers vs. waste more agent calls) is a business decision, not a modeling one, and is carried forward to the recommendation stage.
- Feature importance (Random Forest) confirms poutcome_success as the top predictor, consistent with EDA log row 8. contact_unknown ranks second — flagged as a modeling limitation, since this may reflect a data-recording artifact rather than a genuine behavioral driver, and should not be over-interpreted causally. The engineered previously_contacted flag ranks comparatively low (0.029), suggesting it is partially redundant with previous and poutcome_success.
- No model uses duration, confirming the leakage-prevention decision from preprocessing held throughout model comparison.

Candidates carried forward to evaluation stage: Random Forest and XGBoost, as a near-tied pair with different strengths (general classification quality vs. imbalance-aware ranking quality). Final selection is deferred to the evaluation stage, where threshold selection and business-cost framing (cost of a call vs. value of a missed subscriber) will determine which trade-off the bank should deploy.

Investigation: contact_unknown feature validity

Step	Findings	Evidence
1. Suspicion raised	contact_unknown ranked #2 in Random Forest feature importance (0.1007) — unexpectedly high for a channel-recording gap	Initial feature importance output
2. Temporal clustering test	contact_unknown is almost entirely a proxy for campaign timing, not a genuine "unknown contact channel" signal	96.1% of all unknown-contact records occur in May + June; 84.9% of June contacts and 57.9% of May contacts are labeled unknown, vs. $\leq 8\%$ in every other month
3. Ablation test (XGBoost)	Removing the feature cost a modest, quantified amount of ranking quality	PR-AUC 0.4311 $\rightarrow$ 0.4210 (−2.3% relative)
4. Decision	Dropped contact_unknown from the final feature set across all four models	The feature's predictive value came from duplicating month_may/month_jun through an causally ambiguous path, not from genuine contact-channel information — so it's not worth retaining despite the cost
5. Full re-evaluation impact	Effect is systematic across all models: accuracy and precision rise, recall/ROC-AUC/PR-AUC fall. Decision Tree affected most severely (recall −22% relative); Random Forest and XGBoost most robust to the removal (PR-AUC −2.7% and −2.3% respectively)	See final comparison table below

Final performance comparison (feature set with contact_unknown removed)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.7621	0.2683	0.5983	0.3705	0.7521	0.3945
Decision Tree	0.8533	0.3863	0.4319	0.4079	0.7206	0.3508
Random Forest	0.8474	0.3845	0.5066	0.4372	0.7790	0.4098
XGBoost	0.8327	0.3643	0.5775	0.4468	0.7708	0.4210

Updated conclusion: XGBoost is carried forward as the leading candidate (best F1 and PR-AUC on the final), with Random Forest retained as a second (best ROC-AUC). Decision Tree is retained in the report as the interpretable reference point but is no longer a deployment candidate, given its disproportionate recall loss once the ambiguous feature was removed. Logistic Regression remains useful for its recall/precision trade-off framing but is not the top performer on any ranking-relevant metric.

Investigation: does class-imbalance weighting (scale_pos_weight) help?

Weighted vs. unweighted XGBoost were compared on the final feature set to check whether the effect of imbalance weighting from the other preprocessing decisions.

Configurations	Precisions	Recall	F1	ROC-AUC	PR-AUC
XGBoost (scale_pos_weight = class ratio) → class weight = balanced	0.3643	0.5775	0.4468	0.7708	0.4210
XGBoost (scale_pos_weight = 1, unweighted)	0.6143	0.2514	0.3568	0.7841	0.4270

Interpretation: Weighting does not improve the model's underlying ability to rank clients by subscription likelihood — ROC-AUC and PR-AUC are essentially unchanged (marginally higher without weighting). Weighting's real effect is on the classifier's behavior at the default 0.5 threshold: it more than doubles recall (0.578 vs 0.251) at the cost of precision (0.364 vs 0.614), because the unweighted model defaults toward predicting the majority class ("no") for most clients.

Decision: `scale_pos_weight` is retained in the final model. Since the primary lens is a ranking task (Lift@Top-K, PR-AUC), weighting is not required to improve ranking quality — but it produces a more usable default decision boundary if the bank ever needs a fixed yes/no call list rather than a ranked shortlist, and it avoids handing off a model that defaults to predicting almost no one will subscribe. This is treated as a threshold-usability decision, not a ranking-performance decision, and is stated as such rather than assumed.

Limitation: *This ablation was run on XGBoost only, as the leading candidate, not repeated across all four models. The result is a single-split comparison (no repeated CV), so the small ROC-AUC/PR-AUC differences between weighted and unweighted should not be over-interpreted as statistically significant.*